

HOMEWORK 2
ELEMENTARY LINEAR ALGEBRA
(MATH 2250-03), FALL, 2018

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SECTION 1.4 - MATRIX EQUATIONS

1. Let $A = \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. Determine the subset $\mathcal{B}$ of vectors in $\mathbb{R}^2$ such that the equation $Ax = b$ has a solution if and only if $b \in \mathcal{B}$. Then, give a set of vectors which span $\mathcal{B}$.

Let $A = \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$. We seek the subset $\mathcal{B}$ of vectors in $\mathbb{R}^2$ such that the equation $Ax = b$ has a solution if and only if $b \in \mathcal{B}$. After obtaining $\mathcal{B}$, we also desire a set of vectors which span $\mathcal{B}$. To that end, observe that if $Ax = b$ has a solution, then

$$\begin{aligned} Ax &= b \\ \begin{bmatrix} 2 & -1 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ x_1 \begin{bmatrix} 2 \\ -6 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ -2x_1 \begin{bmatrix} -1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\ (x_2 - 2x_1) \begin{bmatrix} -1 \\ 3 \end{bmatrix} &= \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}. \end{aligned}$$

So, we may conclude that

$$\mathcal{B} = \left\{ (c_2 - 2c_1) \begin{bmatrix} -1 \\ 3 \end{bmatrix} \in \mathbb{R}^2 : c_1, c_2 \in \mathbb{R} \right\}.$$

But then, note that setting $c = c_2 - 2c_1$ and substituting produces

$$\mathcal{B} = \left\{ c \begin{bmatrix} -1 \\ 3 \end{bmatrix} \in \mathbb{R}^2 : c \in \mathbb{R} \right\}.$$

So, by the definition of span, we have

$$\mathcal{B} = \text{span} \left\{ \begin{bmatrix} -1 \\ 3 \end{bmatrix} \right\}.$$

2. Let $v_1 = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ -3 \\ 8 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix}$. Does $\{v_1, v_2, v_3\}$ span $\mathbb{R}^3$? Why or why not?

Let $v_1 = \begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 \\ -3 \\ 8 \end{bmatrix}$ and $v_3 = \begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix}$. We want to determine whether $\text{span}\{v_1, v_2, v_3\} = \mathbb{R}^3$, and justify our claim accordingly. While this would, in general, be solved by creating a matrix $A = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix}$ and discerning whether $Ax = b$ has a solution for all $b \in \mathbb{R}^3$, we may bypass this formalism by observing that

$$-\frac{1}{2}v_1 = \begin{bmatrix} 0 \\ 0 \\ (-1/2)(-2) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad -\frac{1}{3}(v_2 + 4v_1) = \begin{bmatrix} 0 \\ (-1/3)(-3) \\ (-1/3)(8 + (4)(-2)) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\text{and } \frac{1}{4}\left(v_3 - \frac{1}{3}(v_2 + 4v_1) - \frac{5}{2}v_1\right) = \frac{1}{4}\left(\begin{bmatrix} 4 \\ -1 \\ -5 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 5\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

So, as $\text{span}\left\{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\} = \mathbb{R}^3$, we may conclude that $\{v_1, v_2, v_3\}$ span $\mathbb{R}^3$.

In fact, given any $x \in \mathbb{R}^3$, we can directly compute the coefficients $c_1, c_2, c_3 \in \mathbb{R}$ such that

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = x$$

by using the above information as follows:

$$\begin{aligned} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= x_1 \left(\frac{1}{4}\left(v_3 - \frac{1}{3}(v_2 + 4v_1) - \frac{5}{2}v_1\right)\right) + x_2 \left(-\frac{1}{3}(v_2 + 4v_1)\right) + x_3 \left(-\frac{1}{2}v_1\right) \\ &= \frac{-(23x_1 + 32x_2 + 12x_3)}{24}v_1 + \frac{-(x_1 + 4x_2)}{12}v_2 + \frac{x_1}{4}v_3. \end{aligned}$$

So, $c_1 = \frac{-(23x_1 + 32x_2 + 12x_3)}{24}$, $c_2 = \frac{-(x_1 + 4x_2)}{12}$, and $c_3 = \frac{x_1}{4}$.

3. Let $A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix}$. Determine if the columns of A span $\mathbb{R}^4$.

Let $A = \begin{bmatrix} 7 & 2 & -5 & 8 \\ -5 & -3 & 4 & -9 \\ 6 & 10 & -2 & 7 \\ -7 & 9 & 2 & 15 \end{bmatrix}$. We seek to determine if the columns of A span $\mathbb{R}^4$. That is, if we

write the i^{th} column of A as the vector $a_i \in \mathbb{R}^4$ so $A = \begin{bmatrix} a_1 & a_2 & a_3 & a_4 \end{bmatrix}$, i.e

$$a_1 = \begin{bmatrix} 7 \\ -5 \\ 6 \\ -7 \end{bmatrix}, \quad a_2 = \begin{bmatrix} 2 \\ -3 \\ 10 \\ 9 \end{bmatrix}, \quad a_3 = \begin{bmatrix} -5 \\ 4 \\ -2 \\ 2 \end{bmatrix}, \quad \text{and} \quad a_4 = \begin{bmatrix} 8 \\ -9 \\ 7 \\ 15 \end{bmatrix},$$

we seek to determine if $\text{span}\{a_1, a_2, a_3, a_4\} = \mathbb{R}^4$.

To do so, let $b \in \mathbb{R}^4$ be an arbitrary vector, and note that $Ax = b$ is equivalent to

$$x_1 a_1 + x_2 a_2 + x_3 a_3 + x_4 a_4 = b.$$

That is, $Ax = b$ has a solution precisely when $b \in \text{span}\{a_1, a_2, a_3, a_4\}$. So, we may conclude that $\text{span}\{a_1, a_2, a_3, a_4\} = \mathbb{R}^4$ if and only if $Ax = b$ has a solution for all $b \in \mathbb{R}^4$. Now, recall from a theorem given in lecture (or from Theorem 4 in section 1.4 of your book) that $Ax = b$ has a solution for all b if and only if A has a pivot position in every row. As such, routine computation yields that the reduced row echelon form of A is

$$\begin{bmatrix} 1 & 0 & 0 & -93/50 \\ 0 & 1 & 0 & 53/50 \\ 0 & 0 & 1 & -189/50 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

It is then clear that A does not have a pivot position in every row. Therefore, we may conclude that the columns of A do not span $\mathbb{R}^4$.

SECTION 1.5 - SOLUTION SETS OF LINEAR SYSTEMS

4. Find a parametric equation of the line M in $\mathbb{R}^2$ which passes through p and q , where

$$p = \begin{bmatrix} 2 \\ -5 \end{bmatrix} \text{ and } q = \begin{bmatrix} -3 \\ 1 \end{bmatrix}.$$

(Hint: M is parallel to the vector $q - p$.)

We seek a parametric equation of the line M in $\mathbb{R}^2$ which passes through p and q . To that end, we note that the vector

$$q - p = \begin{bmatrix} -3 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -5 \end{bmatrix} = \begin{bmatrix} -5 \\ 6 \end{bmatrix}$$

is parallel to this line. So, using $x \in \mathbb{R}$ as our parameter, M must be of the form $M = x(q - p) + b$ for some vector $b \in \mathbb{R}^2$. That is, if M is parallel to $q - p$, then M is just the line $x(q - p)$ translated up/down (or left/right, if you prefer to think of it that way) by some constant b , which in this case is a vector. To that end, we need only solve for b as we usually would in a regular old college algebra class. That is, since p and q must both lie on this line, then there must be some value $x = x_p \in \mathbb{R}$ at which the line passes through p and some value $x = x_q \in \mathbb{R}$ at which the line passes through q . So, our next step would be solving one of the following equations:

$$p = x_p(q - p) + b \quad \text{or} \quad q = x_q(q - p) + b.$$

For convenience, let's solve $p = x_p(q - p) + b$ (though solving the other equation would work too). Note that this is where *our choice* of parameterization enters into play: We get to *choose* the value of x_p at which this line hits p – we could pick anything we want! For convenience, let's choose $x_p = 0$. This would then produce $p = 0(q - p) + b = b$. So, our line would be given by $M = x(q - p) + p$. Note that this line does indeed pass through q as well – in fact, it does so at $x = 1$, as the following calculation shows:

$$(q - p)(1) + p = q - p + p = q.$$

So, our line is given, in symbols, by $M = x(q - p) + p$ and, in vector notation, by

$$M = x \begin{bmatrix} -5 \\ 6 \end{bmatrix} + \begin{bmatrix} 2 \\ -5 \end{bmatrix}.$$

If we had chosen any other value $x = x_p \in \mathbb{R}$ for the line to pass through p , we would just need to solve $p = x_p(q - p) + b$ for b , producing $b = p - x_p(q - p)$. We can then find our value $x = x_q \in \mathbb{R}$ at which the line passes through q by solving

$$\begin{aligned} q &= x_q(q - p) + b \\ q &= x_q(q - p) + p - x_p(q - p) \\ q &= (x_q - x_p)(q - p) + p. \end{aligned}$$

So, we would just need x_q to satisfy $x_q - x_p = 1$. That is to say, the line passes through q at $x_q = 1 + x_p$, i.e.

$$(1 + x_p)(q - p) + p - x_p(q - p) = q - p + x_p(q - p) - x_p(q - p) + p = q.$$

. In our solution above, we chose $x_p = 0$, so of course we got $x_q = 1$. That is, in general, the line M through q and p which passes through p at $x = x_p \in \mathbb{R}$ must pass through q at $x_q = 1 + x_p$ and has a parametric form given by $M = (x - x_p)(q - p) + p$.

5. Let A be an $n \times n$ matrix, let $b \in \mathbb{R}^n$, and suppose $Ax = b$ has a solution. Explain why this solution is unique precisely when the equation $Ax = 0$ has only the trivial solution $x = 0$.

Let A be an $n \times n$ matrix, let $b \in \mathbb{R}^n$, and suppose $Ax = b$ has a solution. We desire an explanation as to why this solution is unique precisely when the equation $Ax = 0$ has only the trivial solution $x = 0$.

To that end, let's start in the *forward direction* – that is, let's show that if the homogeneous system has a nontrivial solution (i.e. $Ax = 0$ for some $x \neq 0$) and the inhomogeneous system has a solution, then the inhomogeneous system doesn't have a unique solution (i.e. $Ax = b$ has more than one distinct solution). So, let's suppose that $Ax = 0$ has a nontrivial solution $x = v$. That is, let us suppose $Av = 0$ for some vector $v \in \mathbb{R}^n$ such that $v \neq 0$. Then, if $x = w$ is any solution to $Ax = b$, note that

$$A(w + v) = Aw + Av = b + 0 = b.$$

So, if v is a solution to the homogeneous system $Ax = 0$ and w is a solution to $Ax = b$, then $q = v + w$ is also a solution to $Ax = b$. In fact, it is straightforward to see that $y = cv + w$ is also a solution to $Ax = b$ as well for *any* choice of $c \in \mathbb{R}$, as

$$A(cv + w) = A(cv) + Aw = cAv + Aw = c(0) + b = b.$$

Thus, we may conclude that if $Ax = 0$ has a nontrivial solution and $Ax = b$ has any solution at all, then $Ax = b$ has an infinite number of solutions.

Now, let's do the *reverse direction* – that is, let's show that if the inhomogeneous system has more than one solution (i.e. $Ax = b$ has a solution, but it isn't unique), then the homogeneous system has a nontrivial solution. So, let's suppose that $Ax = b$ has two distinct solutions $x = w_1$ and $x = w_2$, i.e. $Aw_1 = Aw_2 = b$ and $w_1 \neq w_2$. Then, note that

$$A(w_1 - w_2) = Aw_1 - Aw_2 = b - b = 0.$$

So, the homogeneous system $Ax = 0$ has $x = w_1 - w_2$ as a solution. Since $w_1 \neq w_2$, it follows that $w_1 - w_2 \neq 0$, so this solution is nontrivial. Thus, we may conclude that if the inhomogeneous system $Ax = b$ has more than one solution, then the homogeneous system $Ax = 0$ has a nontrivial solution.

Note that since we have shown that

$$(\text{Solution to } Ax = b) + (\text{Nontrivial solution to } Ax = 0) \Rightarrow (\text{More than one solution to } Ax = b)$$

and

$$(\text{More than one solution to } Ax = b) \Rightarrow (\text{Solution to } Ax = b) + (\text{Nontrivial solution to } Ax = 0)$$

We have actually shown that these are *equivalent*. That is, the statement " $Ax = 0$ has a nontrivial solution and $Ax = b$ has a solution" implies the statement " $Ax = b$ has more than one solution" and vice versa, so they're really just two different ways of saying *exactly the same thing*!

6. Suppose A is a 3×3 matrix and $y \in \mathbb{R}^3$ is a vector such that the equation $Ax = y$ does not have a solution. Does there exist a vector $z \in \mathbb{R}^3$ such that the equation $Ax = z$ has a unique solution? Why or why not?

Suppose A is a 3×3 matrix and $y \in \mathbb{R}^3$ is a vector such that the equation $Ax = y$ does not have a solution. We want to determine if there exists a vector $z \in \mathbb{R}^3$ such that the equation $Ax = z$ has a unique solution. To that end, let's recall the following theorem from lecture (which is Theorem 4 of section 1.4 in your book):

Theorem 1. Let M be an $m \times n$ matrix. The following statements are equivalent:

- (a) For each $b \in \mathbb{R}^m$, the equation $Mx = b$ has a solution.
- (b) Each $b \in \mathbb{R}^m$ is a linear combination of the columns of M
- (c) The columns of M span $\mathbb{R}^m$
- (d) M has a pivot position in every row.

So, if $Ax = y$ does not have a solution for some $y \in \mathbb{R}^3$ (like we initially supposed), then it is apparent that $Ax = b$ does not have a solution for every $b \in \mathbb{R}^3$. Thus, A does not have a pivot position in every row by the above theorem. So, since A has the same number of columns as it has rows, this tells us that A cannot have 3 pivot columns. That is to say, the system $Ax = 0$ has at least one free variable. So, the homogeneous system $Ax = 0$ must have nontrivial solutions. As we demonstrated in Problem 5, this then implies that $Ax = z$ cannot have a unique solution for any $z \in \mathbb{R}^3$.

Hence, if A is a 3×3 matrix and $Ax = y$ does not have a solution for some $y \in \mathbb{R}^3$, we may conclude that there does not exist a vector $z \in \mathbb{R}^3$ such that the inhomogenous system $Ax = z$ has a unique solution.

As a side note, it is easy to check that the above argument goes through just fine if A is an $n \times n$ matrix instead of the particular case where A is a 3×3 matrix. So, this is really a statement about square matrices (i.e. matrices with the same number of rows and columns). That is, if A is a square matrix of any size and $Ax = y$ doesn't have a solution for some y , then $Ax = z$ cannot have a unique solution for any z .

SECTION 1.7 - LINEAR INDEPENDENCE

7. Find the value(s) of h for which the vectors $v_1 = \begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} -5 \\ 7 \\ 8 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 1 \\ h \end{bmatrix}$ are linearly dependent in $\mathbb{R}^3$.

We seek the value or values of $h \in \mathbb{R}$ for which the set of vectors $\{v_1, v_2, v_3\}$ given above are linearly dependent. That is, using the definition of linear dependence, we seek the values of h for which the vector equation

$$x_1 v_1 + x_2 v_2 + x_3 v_3 = 0$$

has a solution with at least one of the x_i 's not equal to zero. That is, we seek the values of h for which the homogenous system

$$\begin{bmatrix} 1 & -5 & 1 \\ -1 & 7 & 1 \\ 3 & 8 & h \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

has a nontrivial solution. To that end, we form the augmented matrix for this system, which is

$$\begin{bmatrix} 1 & -5 & 1 & 0 \\ -1 & 7 & 1 & 0 \\ 3 & 8 & h & 0 \end{bmatrix}$$

Following this, routine calculation yields the following (almost-reduced) row echelon form of the augmented matrix for our system

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 26-h & 0 \end{bmatrix}.$$

So, if $h = 26$, then the third column and third row of the augmented matrix consist only of zeroes, and the augmented matrix will not have a pivot in every row. That is, if $h = 26$, then the system above will have a free variable. So, using the fact that a homogenous system $Ax = 0$ has a nontrivial solution if and only if the system $Ax = 0$ has a free variable, it follows that the set $\{v_1, v_2, v_3\}$ is linearly dependent if and only if $h = 26$.

8. Let $A = \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & 3 \\ 9 & -3 & 3 \end{bmatrix}$. Solve $Ax = 0$ without performing any row operations. Explain how you obtained this answer and justify your reasoning.

We seek a solution to $Ax = 0$, where A is as given above. To that end, note that the third column of A is equal to the first column plus twice the second column, i.e.

$$\begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix}.$$

So, since the matrix-vector product Ax is just a linear combination of the columns of A , we then have

$$\begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & 3 \\ 9 & -3 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} - \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

In fact, for any choice of $x \in \mathbb{R}$, we have

$$\begin{aligned} \begin{bmatrix} 4 & 1 & 6 \\ -7 & 5 & 3 \\ 9 & -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ 2x \\ -x \end{bmatrix} &= x \begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix} + 2x \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} - x \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} \\ &= x \left(\begin{bmatrix} 4 \\ -7 \\ 9 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix} - \begin{bmatrix} 6 \\ 3 \\ 3 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \end{aligned}$$

9. Suppose A is an $m \times n$ matrix such that the equation $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$. Must the columns of A be linearly independent? Why or why not?

Suppose A is an $m \times n$ matrix such that the equation $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$. We seek to determine whether the columns of A must be linearly independent. To do so, we will use a technique called *proof by contradiction*. That is, we'll suppose that the columns of A are not linearly independent, then we'll show that this *contradicts* our assumption that $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$. This tells us that the statements " $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$ " and " A has linearly dependent columns" cannot both, simultaneously, be true. So, if we assume $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$, it must be the case that A does *not* have linearly dependent columns, which is the same as saying that A has linearly *independent* columns.

Suppose A is an $m \times n$ matrix and suppose $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$. Suppose as well that the columns of A are linearly dependent, and denote the i^{th} column of A by a_i (i.e. $A = \begin{bmatrix} a_1 & \dots & a_m \end{bmatrix}$ with $a_i \in \mathbb{R}^n$). Then, there exists constants $c_1, \dots, c_m \in \mathbb{R}$ with at least one $c_i \neq 0$ such that $c_1 a_1 + \dots + c_m a_m = 0$. Equivalently, since

$$Ax = \begin{bmatrix} a_1 & \dots & a_m \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_m \end{bmatrix} = x_1 a_1 + \dots + x_m a_m,$$

there exists a vector $v \in \mathbb{R}^m$ with $v \neq 0$ such that $Av = 0$ – for example, take $v = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$. That is,

the homogenous system $Ax = 0$ has a nontrivial solution. Hence, we may see that $Ax = 0$ holds for both $x = v$ and $x = 0$. Thus, we have shown that $Ax = b$ does not have a unique solution for all $b \in \mathbb{R}^n$, which contradicts our original assumption to the contrary.

As such, we may conclude that if $Ax = b$ has at most one solution for all $b \in \mathbb{R}^n$, then A must have linearly independent columns.